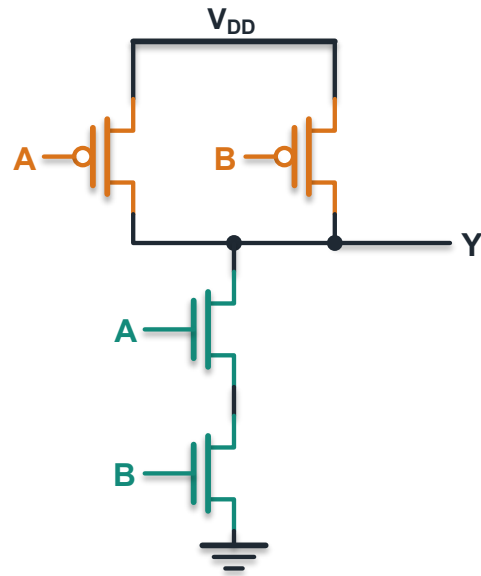


Inside the Logic Gate

How NOT, AND, OR, NAND, NOR and XOR are built from transistors — and why they behave the way they do



A CMOS NAND gate: 4 transistors

A logic gate is a circuit of switches

1

Logic levels are voltages

1 = HIGH $\approx$ VDD (e.g., 3.3 V or 5 V). 0 = LOW $\approx$ 0 V (GND).

2

A MOSFET is a switch

Its gate voltage opens or closes a conducting channel between the other two terminals (source and drain).

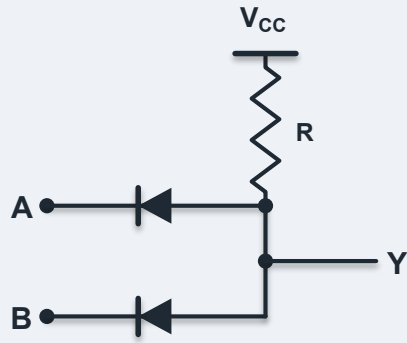
3

Two complementary types

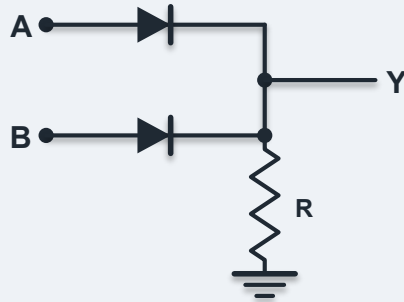
NMOS turns ON when its gate is 1 and pulls strongly to 0. PMOS (bubble on the gate) turns ON when its gate is 0 and pulls strongly to 1.



The simplest electrical gates: diode logic



Diode AND



Diode OR

Diode AND

If any input is 0 V, its diode conducts and drags Y low (≈ 0.7 V). Only when all inputs are high does R pull Y up to V_{CC} .

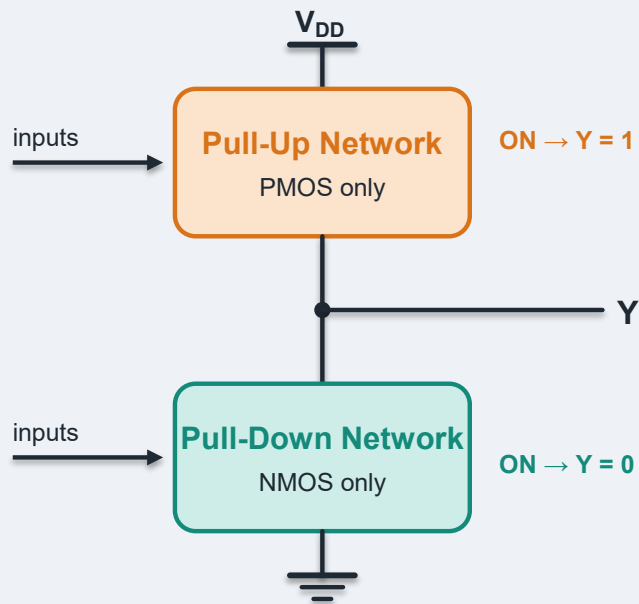
Diode OR

Any high input forward-biases its diode and drives Y high ($\approx V_{IN} - 0.7$ V). With all inputs low, R pulls Y to GND.

Why we moved on

No inversion (no NOT, NAND, NOR), ~ 0.7 V lost per stage, and no gain to drive other gates. Transistors fix all three.

The CMOS recipe for every gate



Pull-up network

PMOS between V_{DD} and Y . Conducts exactly when Y should be 1.

Pull-down network

NMOS between Y and GND. Conducts exactly when Y should be 0.

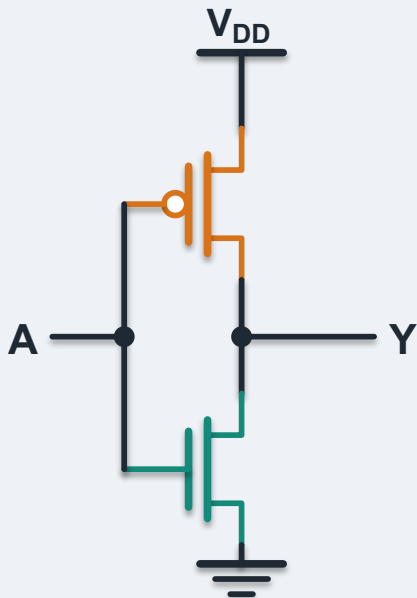
Duality

Series NMOS $\leftrightarrow$ parallel PMOS (and vice versa). Only one network is ever ON.

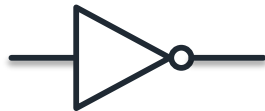
Always inverting

One CMOS stage produces NOT, NAND or NOR. AND and OR need a second stage.

NOT (Inverter)



■ PMOS pull-up ■ NMOS pull-down



$$Y = A'$$

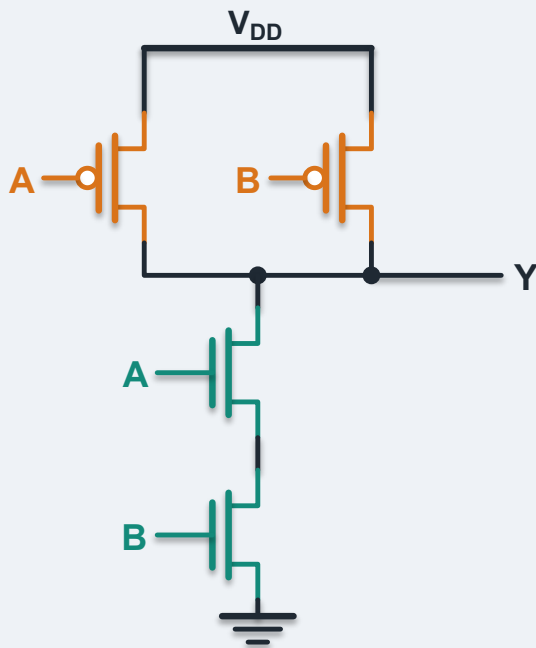
A	Y
0	1
1	0

- A = 0: PMOS ON, NMOS OFF → Y connects to VDD (1).
- A = 1: NMOS ON, PMOS OFF → Y pulls to GND (0).
- Only one transistor conducts at a time, so no steady current flows VDD → GND.
- Every other CMOS gate is a generalization of this circuit.

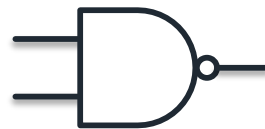
NAND

4

transistors



■ PMOS pull-up ■ NMOS pull-down

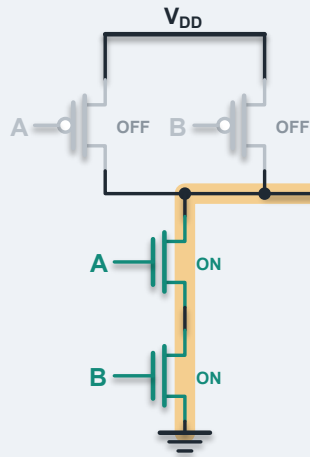


$$Y = (A \cdot B)'$$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

- Pull-down: NMOS in series. Y goes LOW only if A AND B are both 1.
- Pull-up: PMOS in parallel. Y goes HIGH if A OR B is 0.
- Series $\leftrightarrow$ parallel duality: exactly one network conducts.
- Universal gate: any logic function can be built from NANDs alone.

Tracing current through a NAND

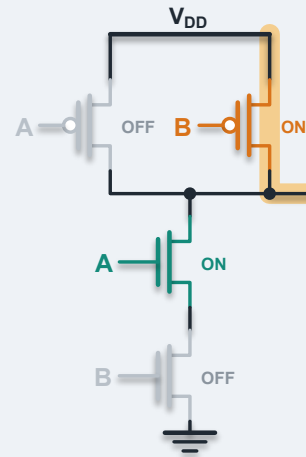


$A = 1, B = 1 \rightarrow$
 $Y = 0$

Both NMOS are ON, forming a path from Y to GND. Both PMOS are OFF, so VDD is disconnected.

— highlighted = conducting path

Grey transistors are OFF (open switches).



$A = 1, B = 0 \rightarrow$
 $Y = 1$

PMOS B is ON and connects Y to VDD. NMOS B is OFF, which breaks the series path to GND.

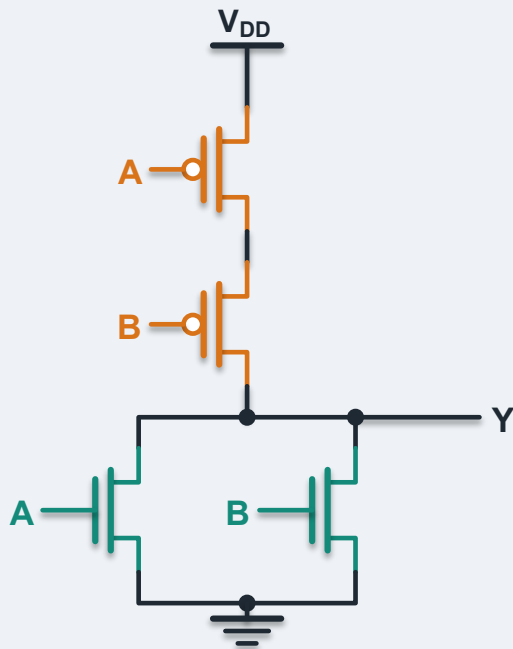
— highlighted = conducting path

Grey transistors are OFF (open switches).

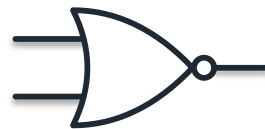
NOR



transistors



■ PMOS pull-up ■ NMOS pull-down

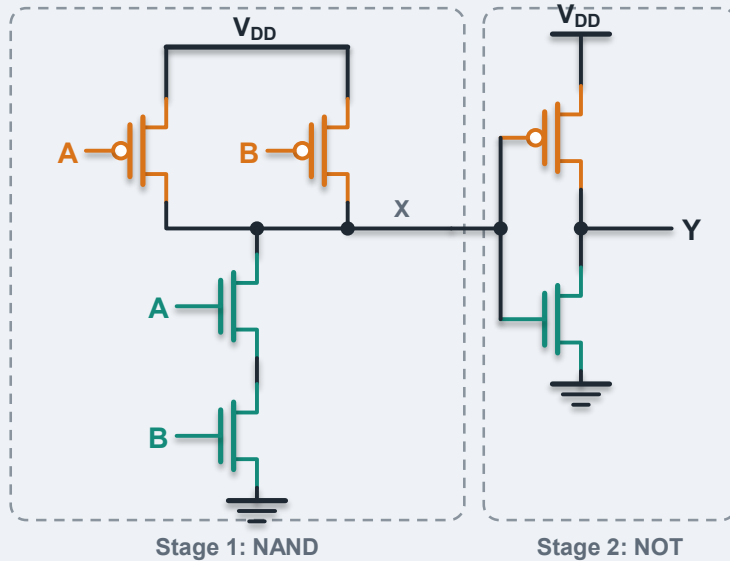


$$Y = (A+B)'$$

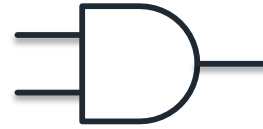
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

- Pull-down: NMOS in parallel. Either input at 1 pulls Y LOW.
- Pull-up: PMOS in series. Y is HIGH only when A = 0 AND B = 0.
- Series PMOS are slow (holes are ~2–3× less mobile than electrons), so NOR is slower than NAND.
- Also a universal gate.

AND = NAND + NOT



■ PMOS pull-up ■ NMOS pull-down



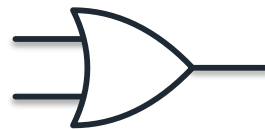
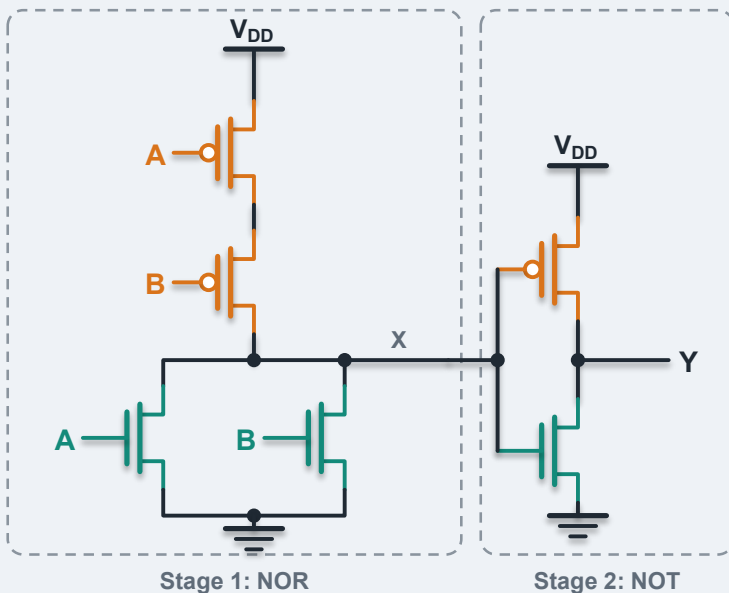
$$Y = A \cdot B$$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

- One CMOS stage can't be non-inverting: NMOS pass a weak 1 and PMOS a weak 0.
- So invert twice: $X = (AB)'$, then $Y = X' = AB$.
- Cost: $4 + 2 = 6$ transistors and two gate delays.
- The inverter also restores a strong output.

OR = NOR + NOT

6
transistors

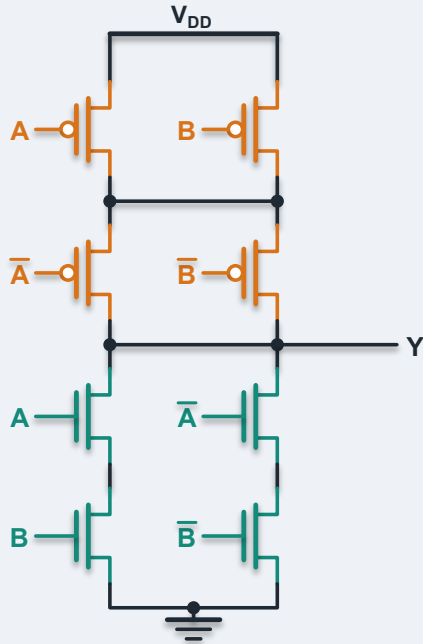


$$Y = A + B$$

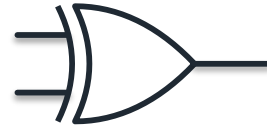
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

- $X = (A+B)'$ from the NOR, then $Y = X' = A + B$.
- $4 + 2 = 6$ transistors, two stages of delay.
- The series PMOS in the NOR stage make OR slower than AND.
- Designers use De Morgan to absorb inversions and use NAND/NOR directly.

XOR (Exclusive OR)



■ PMOS pull-up ■ NMOS pull-down

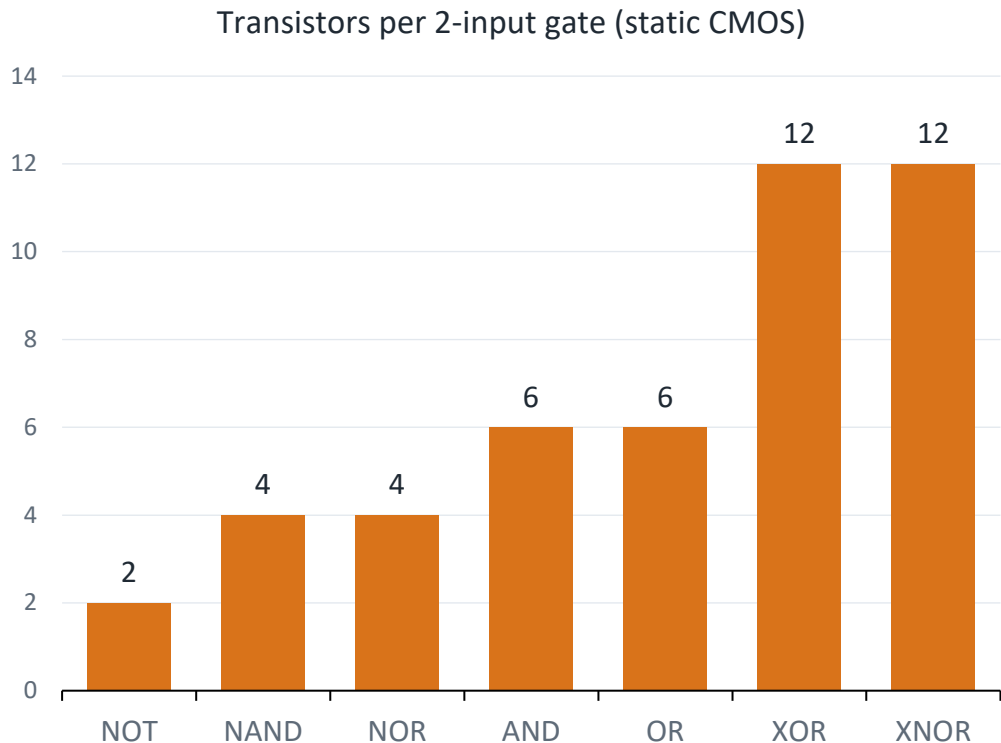


$$Y = A \oplus B$$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

- Write as one inverting stage: $Y = (AB + A'B')'$.
- Pull-down: (A series B) $\parallel$ (A' series B'). Y goes LOW when inputs match.
- Pull-up is the dual: (A $\parallel$ B) in series with ($A' \parallel B'$).
- A' and B' come from two inverters (+4) $\rightarrow$ 12 total. Transmission-gate designs use 6–8.

Cost of each gate in CMOS



Inverting gates are native

NOT, NAND and NOR need only one stage: less area, delay and power.

NAND beats NOR

Series NMOS (fast electrons) hurt less than series PMOS (slow holes).

XOR is expensive

Needs complemented inputs, so it is often built with transmission gates instead.

What the circuit view tells us

$$\approx 0$$

Static power

With steady inputs there is no DC path from VDD to GND — one network is always open.

$$\propto CV^2f$$

Dynamic power

Energy is spent charging and discharging the load capacitance every time Y switches.

$$R \cdot C$$

Propagation delay

On-resistance of the conducting path $\times$ load capacitance. Series stacks add resistance.

$$0 \leftrightarrow VDD$$

Rail-to-rail output

Outputs swing fully to the supplies, giving wide noise margins (V_{IL} , V_{IH}).

Key takeaways

- 1 Every gate is a PMOS pull-up network plus an NMOS pull-down network wired as duals.
- 2 Series transistors act like AND, parallel transistors act like OR.
- 3 NOT, NAND and NOR are single-stage; AND and OR add an inverter; XOR needs complemented inputs.
- 4 Transistor count and series stacking drive a gate's area, speed and power.